

ETF Allocation Decision Support

A Business Intelligence Project Using Tableau, Python, and dbdiagram.io

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Abstract

This report presents a business intelligence project designed to support an investment decision problem: selecting and allocating capital across regional and style-based equity ETFs. The decision maker is an investment analyst or asset allocation committee. The project combines ETF price history, ETF characteristics, country and industry exposures, macroeconomic indicators, and Python-generated portfolio optimization outputs. Tableau is used for interactive dashboards and storytelling, dbdiagram.io is used for data modeling, and Python is used for data preparation, financial metric calculation, and Markowitz-style portfolio optimization. The final output is a structured Tableau Story that guides the decision maker from ETF screening to final optimized portfolio weights.

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1 Introduction

1.1 Problem addressed by the project

In this project, I address an investment decision problem that is frequently faced in the financial sector: how to allocate funds across assets while taking different criteria into account. I focus on equity-based ETFs, or exchange-traded funds, because they provide rich information beyond prices. More specifically, ETFs may have a regional focus, such as the United States, Europe, Japan, China, India, or Emerging Markets; a style focus, such as value, growth, dividend, momentum, large-cap, or small-cap; a sector focus, such as technology, healthcare, financials, or energy; and an index choice, meaning whether they are actively managed or follow a specific index.

These dimensions are important because the final portfolio should not only contain high-performing ETFs, but should also provide meaningful international allocation. The decision maker is an investment analyst at a wealth management firm or an asset allocation committee responsible for building an international equity portfolio.

The main decision supported by the project is:

Which regional and style-based ETFs should form a balanced international equity portfolio?

The ETF universe is large and difficult to compare manually. Many ETFs may look attractive at first sight, but they can be quite different in terms of risk, geography, sector exposure, cost, liquidity, and macroeconomic background. For example, one ETF may have strong historical returns but high volatility. Another ETF may be cheap but not very diversified. A third ETF may appear attractive by name, but it may be strongly exposed to countries or industries facing weaker macroeconomic conditions.

Therefore, the analyst should not select ETFs only by looking at past returns. The decision requires a broader view. That is why the project combines financial performance, risk measures, geographic exposure, industry exposure, style classification, macroeconomic indicators, and portfolio allocation metrics. This makes the decision more structured and more transparent.

The decision process has five main steps. First, the analyst screens the ETF universe and excludes ETFs that do not match the relevant investment scope. Second, the analyst examines each ETF's actual exposure in terms of industries, regions, and countries. Third, the analyst checks the macroeconomic context behind the ETF's geographic exposure. Fourth, the analyst compares ETFs by return, volatility, Sharpe ratio, drawdowns, correlations, and other risk metrics. Finally, the analyst uses portfolio optimization to recommend a set of ETF weights and display important metrics of the resulting portfolio.

The final recommendation is based on Markowitz portfolio theory. In this logic, ETFs are evaluated both individually and as part of a portfolio. The project uses expected return, volatility, covariance, correlation, and Sharpe ratio to analyze the risk-return trade-off. In the final dashboard, portfolio optimization is used to build an efficient frontier of possible ETF portfolios. The

recommended portfolio is selected as the portfolio with the maximum Sharpe ratio, given practical constraints such as long-only weights, maximum ETF weight, and the risk-free rate. This follows the mean-variance logic introduced by [Markowitz \(1952\)](#).

1.2 Data used to support the decision

The project uses real data from several sources and combines them into one analytical framework. ETF price data is used to calculate financial performance and risk metrics, such as monthly returns, cumulative returns, volatility, Sharpe ratio, momentum, drawdown, and correlations. This data is important because an analyst needs to understand how ETFs behaved historically and how risky they were. ETF historical prices were retrieved from Yahoo Finance, which provides historical market data for securities and ETFs ([Yahoo Finance, 2026](#)).

The project also includes ETF characteristics: assets under management, expense ratio, bid-ask spread, geographic scope, development group, style focus, size focus, sector focus, and whether the ETF is active or index-based. These characteristics describe what type of ETF the analyst is considering and whether it fits the investment goals. ETF characteristics, holdings, and exposures were collected from iShares by BlackRock ([BlackRock iShares, 2026](#)).

Country and industry exposure data are also important. An ETF is not just a ticker symbol, but a basket of countries, regions, sectors, and industries. Exposure data helps the decision maker understand what is actually inside the ETF. For example, two ETFs may both be international equity ETFs, but one may be more exposed to Europe, while another may be more exposed to Asia-Pacific or Emerging Markets.

Macroeconomic data is included as an explanatory layer. Indicators such as inflation, unemployment, interest rates, and exchange rates help the analyst interpret the economic environment of the countries represented in an ETF. These indicators do not directly determine the final portfolio. Rather, they help explain why some regions or styles may be more attractive or risky under current conditions. The macroeconomic indicators were retrieved from FRED, the Federal Reserve Economic Data platform ([FRED, 2026](#)).

Python is used to create additional derived tables, such as return metrics, risk metrics, drawdowns, correlations, and optimizer outputs. These prepared tables are then imported into Tableau for visualization and storytelling.

1.3 Ex-ante justification for the tools

Tableau is a popular business intelligence tool. It is appropriate for this project because the problem requires interactive analysis and clear communication with a decision maker. The analyst needs to filter ETFs by different criteria, compare risk and return, inspect exposures, analyze time-related trends, and move step by step toward a final recommendation. Tableau supports this process through charts, filters, maps, scatter plots, bar charts, heatmaps, dashboards, and stories.

Tableau is also useful because the final output must be understandable for a non-technical decision

maker. The project is not only about calculating financial metrics. It also needs to explain the investment logic behind the allocation. A Tableau Story is therefore suitable because it guides the reader through the full decision process: starting from the business task and data foundation, then moving to ETF screening, exposure analysis, macroeconomic context, risk analysis, and the final recommended portfolio allocation.

dbdiagram.io is used for data modeling, as the project combines several related tables and requires a clear data structure before visualization. The tool allows users to draw database relationship diagrams using a simple DSL language (dbdiagram.io, 2026). The schema follows a galaxy schema logic. It includes dimensions such as ETF, month, geography, industry, benchmark, and macro indicator. Around these dimensions, several fact tables store analytical measures: ETF monthly market performance, ETF metrics, geographic exposure, industry exposure, and monthly macroeconomic observations. The main schema is included in Appendix A.

Python is a core element for advanced data preparation and portfolio computation. Since the data comes from different sources, it is easier and more reliable to prepare it using Python before importing it into Tableau. Python is used to clean dates and ticker names, convert daily prices to monthly data, calculate returns, log returns, volatility, Sharpe ratio, drawdowns, and momentum. It is also used to prepare the optimizer tables and implement the Markowitz portfolio optimization logic.

Thus, the combination of Tableau, dbdiagram.io, and Python provides strong capabilities for conducting all parts of the process: from collecting and preparing the data, to documenting the model and turning it into an interactive decision-support story.

2 Data modeling and preparation

2.1 Data model

The data model was designed as a galaxy schema in dbdiagram.io. This structure was chosen because the project combines several analytical views: ETF performance, ETF characteristics, geographic exposure, industry exposure, macroeconomic indicators, and portfolio optimization. A single flat table would not be appropriate because the same ETF must be analyzed across time, countries, regions, industries, and macroeconomic indicators.

The main dimension table is **etf**. It stores the ETF identifier, ticker, ETF name, provider, asset class, geographic scope, development group, style focus, size focus, sector focus, and active versus index classification. This table is connected to the **benchmark** table, which stores the benchmark index followed by each ETF. The **month** table supports time-based analysis by month, quarter, and year. The **geography** and **industry** tables support exposure analysis by country, parent region, industry label, and industry cluster. The **macro_indicator** table stores the macroeconomic indicator names and their default units.

The main fact table for market performance is **etf_market_monthly**. It stores monthly prices, adjusted close prices, volume, trading days, monthly returns, log returns, 12-month momentum,

and cumulative return index. This table supports performance analysis, return comparison, momentum analysis, and time-series visualizations in Tableau.

Another important table is `etf_metric`. This table stores ETF-level characteristics and financial metrics, such as expense ratio, net assets, bid-ask spread, number of holdings, P/E ratio, P/B ratio, three-year standard deviation, and equity beta. These fields are used mainly for ETF screening and comparison. They help the decision maker evaluate not only return, but also cost, liquidity, valuation, and risk.

The model also includes two exposure fact tables: `etf_geography_exposure` and `etf_industry_exposure`. These tables connect ETFs to countries and industries through exposure weights. They make it possible to analyze what each ETF actually contains. This is important because the project does not treat ETFs only as tickers. It also analyzes their underlying geographic and industry composition.

Macroeconomic data is modeled through `macro_series` and `macro_observation_monthly`. The `macro_series` table connects a macroeconomic indicator to a geography and defines its frequency, source, and unit. The `macro_observation_monthly` table stores the actual monthly macro values. This structure supports analysis by country, region, macro indicator, and time.

Overall, the schema supports analysis by ETF, month, country, region, industry, benchmark, and macro indicator. This fits the multi-dimensional nature of the investment decision. The same model supports the full Tableau story: ETF screening, exposure analysis, macro context, risk comparison, and final portfolio recommendation. The schema image is provided in Appendix 1.

For the final recommendation dashboard, an additional Tableau-ready data source was created from Python optimization outputs. This second data source includes scenario, efficient frontier, optimized weights, KPI, and final exposure tables. It is separate from the main galaxy schema because the optimizer output has a different analytical grain: it describes simulated portfolios and the final recommended portfolio, rather than raw ETF observations. The optimizer schema is shown in Appendix 2.

2.2 Data preparation

Python was used as the main data preparation tool. The pipeline started from three main raw data sources: ETF metadata and exposure data from iShares / BlackRock, historical ETF prices from Yahoo Finance, and macroeconomic indicators from FRED.

First, ETF metadata and exposure information were collected from iShares / BlackRock and cleaned. This step produced the ETF universe, key fund characteristics, benchmark information, cost fields, and country and industry exposure tables. The data was standardized so that the same types of values could be used consistently in Tableau filters.

Second, daily ETF prices from Yahoo Finance were converted into monthly data. From these prices, Python calculated monthly returns, log returns, cumulative return index, 12-month momentum, running peak index, and drawdown. These calculations were necessary because Tableau dashboards required ready-to-use financial metrics, not only raw daily prices.

Third, macroeconomic data from FRED was cleaned and transformed into a monthly format. Duplicate rows were removed, country names were standardized, and only observations relevant for Tableau analysis were kept. This allowed macro indicators to be connected to countries and compared over time.

Fourth, ETF classification fields were standardized. These fields include geographic scope, developed or emerging group, style focus, size focus, sector focus, and active versus index classification. This step was important because Tableau filters need clean and understandable categories. To preserve as much data as possible, an optional OpenAI-based step was used only to refine some difficult classifications that were not fully captured by the parser. The final classification values were still reviewed and standardized before use in Tableau.

Finally, all cleaned tables were exported into Tableau-ready Excel workbooks. The main workbook contains the galaxy schema tables, while the second workbook contains the portfolio optimization output.

2.3 Portfolio optimization

Portfolio optimization was prepared in Python as a separate analytical output. It is based on Markowitz portfolio theory, which evaluates portfolios by comparing expected return and risk. In this framework, risk is measured by volatility, and diversification is captured through the covariance and correlation structure between ETFs ([Markowitz, 1952](#)).

The portfolio return is calculated as the weighted average of ETF expected returns. Portfolio risk is calculated using the covariance matrix of ETF returns. In this project, monthly ETF returns were used to simulate many possible long-only portfolios. Each portfolio had practical constraints: weights had to be non-negative, weights had to sum to one, and each ETF had a maximum allowed portfolio weight.

For each simulated portfolio, Python calculated expected return, expected volatility, Sharpe ratio, and maximum drawdown. The Sharpe ratio measures risk-adjusted performance. It is calculated as expected portfolio return minus the risk-free rate, divided by portfolio volatility. A higher Sharpe ratio means that the portfolio offers more return per unit of risk.

The final portfolio was selected as the feasible portfolio with the highest Sharpe ratio. This means that the recommendation is not based only on the highest expected return, but on the best risk-adjusted return among the simulated portfolios. The optimization results were then exported into a separate Tableau-ready workbook. This workbook includes the efficient frontier, selected ETF weights, portfolio KPIs, and final country and industry exposure of the optimized portfolio.

2.4 Tableau-ready data

To sum up, the final output consists of two Tableau-ready workbooks. The first workbook contains the main data model with ETF, market, metric, exposure, geography, industry, macro, and time tables. This workbook supports ETF screening, exposure analysis, macro context, and risk

comparison.

The second workbook contains the portfolio optimization output. It includes tables for optimization scenarios, efficient frontier points, optimized ETF weights, portfolio KPIs, and final portfolio exposure. This second data source supports the final recommendation dashboard.

This preparation step was important because Tableau works best when the data is already structured clearly. Python was used for cleaning, transformation, metric calculation, and optimization, while Tableau was used for interactive analysis, dashboards, and storytelling.

3 Application

The application part of the project is presented through one Tableau Story. The goal of the story is to guide the decision maker from the initial ETF universe to the final portfolio allocation. Each story point answers one part of the main decision problem: which regional and style-based ETFs should form a balanced international equity portfolio? Screenshots of the main story scenario are included in Appendix B. A second scenario is included in Appendix C.

The first story point introduces the business task and the data foundation. It explains that the decision maker is an investment analyst or an asset allocation committee. It also shows that the project combines ETF price data, ETF characteristics, country and industry exposures, macroeconomic indicators, and Python-generated portfolio metrics. This creates the context for the rest of the story and makes clear that the project is not only descriptive, but decision-oriented.

The second story point is the ETF Screener dashboard. Its role is to reduce the full ETF universe to a smaller set of suitable candidates. The analyst can filter ETFs by geographic scope, development group, style focus, size focus, sector focus, and active versus index classification. The dashboard also includes practical screening measures such as assets under management, expense ratio, bid-ask spread, and return history. This supports the first important decision: which ETFs should be kept for deeper analysis, and which ETFs should be excluded. The first dashboard is shown in Appendix 3.

The third story point is the Exposure dashboard. It answers the question: what is the investor actually buying? The dashboard shows the country, regional, and industry exposure of the selected ETF. This is important because ETFs are not only ticker symbols; they are baskets of countries, sectors, and companies. This dashboard helps the analyst detect possible concentration, unintended exposure, or mismatch between the ETF name and its actual composition.

The fourth story point is the Macro Context dashboard. It connects ETF exposure with macroeconomic indicators such as inflation, unemployment, interest rates, and exchange rates. This dashboard does not directly produce portfolio weights, but it adds an economic interpretation layer to the investment decision. For example, if an ETF is strongly exposed to a few countries, the analyst can check whether these countries show stable or risky macroeconomic conditions over time.

The fifth story point is the Risk and Diversification dashboard. It compares the selected ETFs

by annualized return, volatility, Sharpe ratio, maximum drawdown, and correlations. This step is important because a good portfolio should not only include ETFs with strong individual performance. It should also include ETFs that diversify each other. The correlation matrix helps show whether the selected ETFs bring real diversification benefits or whether they move too similarly.

The final analytical story point is the Portfolio Recommendation dashboard. This dashboard uses Python-generated optimizer tables based on Markowitz portfolio theory. The optimizer builds an efficient frontier of possible ETF portfolios and selects the feasible portfolio with the maximum Sharpe ratio, given constraints such as long-only weights and a maximum ETF weight. The dashboard shows the optimized ETF weights, expected return, expected volatility, Sharpe ratio, maximum drawdown, and final country and industry exposure. This turns the previous analysis into a concrete portfolio recommendation.

The optimization process was made reusable through a Python command-line workflow. The user chooses the final ETF tickers to include and then runs:

```
poetry run bi-build-portfolio-optimizer --max-weight 0.3 \  
--risk-free-rate 0.03 --etfs-list "DIVB,DVY,DVYA,HDV,IDV,IVVW,PFF"
```

Here, `max-weight` is the maximum allowed weight for one ETF, `risk-free-rate` is used to calculate the Sharpe ratio, and `etfs-list` defines the candidate ETFs for optimization. The script applies these assumptions, generates the optimizer tables, and exports them for Tableau. The only remaining step is to refresh the Tableau data source, after which the final recommendation dashboard updates automatically. More details are available in the README file accompanying the code.

Tableau is used for interactive analysis and storytelling, while Python supports the more advanced calculations. Python prepares financial metrics, drawdowns, correlations, efficient frontier points, optimized weights, and portfolio KPIs. These outputs are exported to Excel and used in Tableau as additional data sources. This combination allows Tableau to remain focused on decision support and visualization, while Python handles the calculations that are more complex to perform directly inside Tableau.

To demonstrate that the recommendation depends on the analyst's assumptions, I also prepared two portfolio scenarios, which are shown in the appendix. They apply different screening filters and therefore produce different optimized ETF weights. This shows that the BI tool does not give one fixed answer. Instead, it supports decision making under different investment views and constraints.

Overall, the Tableau Story supports the decision problem in a structured way. It first screens ETFs, then explains their exposures, checks macro context, compares risk and diversification, and finally recommends an optimized portfolio. The main insight is that ETF allocation should not be based only on past returns. A stronger decision requires combining performance, risk, cost, exposure, macroeconomic context, and portfolio optimization.

4 Conclusion

Business intelligence tools, supported by Python, provided strong support for solving the ETF allocation problem. The main value of the BI approach was that it transformed a complex investment decision into a clear analytical process with understandable visual outputs. Instead of looking only at isolated return numbers, the Tableau Story made it possible to analyze the ETF universe step by step: screening candidates, checking exposures, reviewing macroeconomic context, comparing risk and diversification, and finally recommending optimized portfolio weights.

Ex post, the choice of tools proved to be appropriate. Python was useful for data cleaning, financial metric calculation, correlation analysis, drawdown calculation, and portfolio optimization. These tasks would have been difficult and less transparent to perform only inside Tableau. db-diagram.io helped document the galaxy schema and clearly show the relationships between ETF, time, geography, industry, macro, and optimization tables. Tableau was effective for building interactive dashboards and presenting the final recommendation to a decision maker in a structured way.

The data was also suitable for the decision problem. The project combined ETF price history, ETF characteristics, country and industry exposures, and macroeconomic indicators. This made the analysis richer than a simple ranking based on historical return. The decision maker could compare ETFs not only by performance, but also by cost, liquidity, risk, exposure, macro context, and diversification potential.

The main lesson from the project is that data preparation and data modeling are central to any BI project. Good dashboards depend on clean data, consistent definitions, and a well-structured model. Without this foundation, the final visualizations may look attractive but would not support reliable decision making. The project also showed that business intelligence is not only about visualization. It is about connecting data, models, calculations, and storytelling to support a real decision.

At the same time, the project revealed some limitations of Tableau. Tableau is strong for interactive visualization, but it is less flexible for advanced data operations. For example, building correlation matrices, applying complex filters across several related tables, or updating portfolio optimization dynamically is easier in Python than directly inside Tableau. This is why the project used Python as a complementary tool for advanced calculations.

If I were to develop the project further, I would use Tableau-Python integration to make portfolio optimization more dynamic. This would allow the user to change the ETF list or constraints directly from Tableau and update the optimized weights without switching to the Python terminal. I would also try to collect data from more ETF providers, not only iShares / BlackRock, to make the ETF universe broader and less dependent on one source.

If I were to start again, I would spend even more time designing the data model before building dashboards. I would also use agentic AI more systematically to accelerate data preprocessing, classification, and documentation. However, I would still keep human review in the process, especially for financial classifications and portfolio interpretation. Overall, the project showed

that BI tools can meaningfully support investment decision making when they are combined with careful data preparation, clear modeling, and a structured decision story.

References

- BlackRock iShares. (2026). *iShares ETFs by BlackRock*. Retrieved in June 2026 from <https://www.ishares.com/us>
- dbdiagram.io. (2026). *Database Relationship Diagrams Design Tool*. Retrieved in June 2026 from <https://dbdiagram.io/>
- Federal Reserve Bank of St. Louis. (2026). *Federal Reserve Economic Data (FRED)*. Retrieved in June 2026 from <https://fred.stlouisfed.org/>
- Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>
- Yahoo Finance. (2026). *Historical data for financial securities*. Retrieved in June 2026 from <https://finance.yahoo.com/>

A Data schemas

This appendix contains the data models created in dbdiagram.io. The first schema shows the main galaxy schema used for ETF, market, exposure, macro, and time-related analysis. The second schema shows the optimizer output structure used for the final recommendation dashboard.

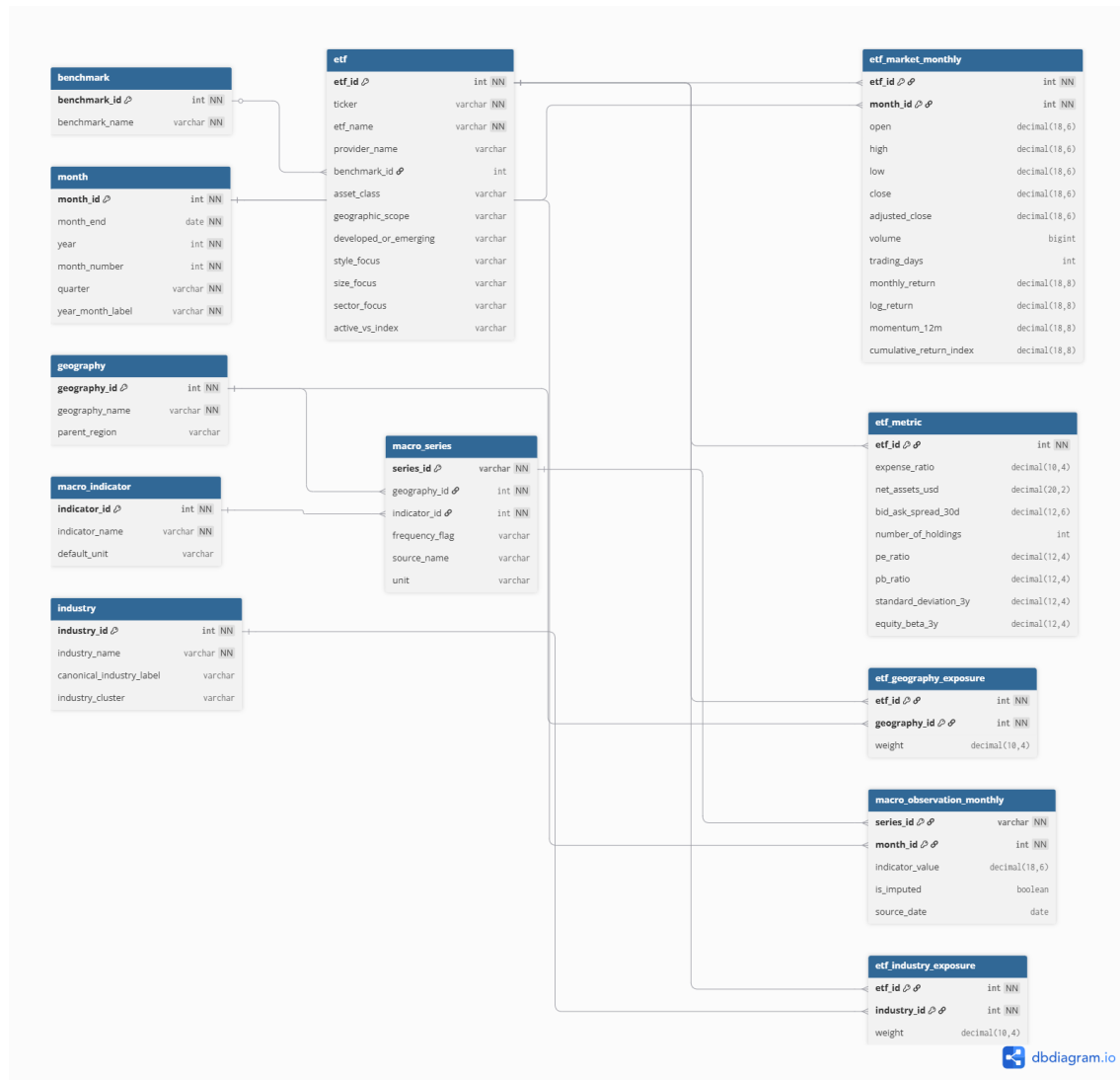


Figure 1: Main galaxy schema for ETF, market, exposure, macro, and time tables.

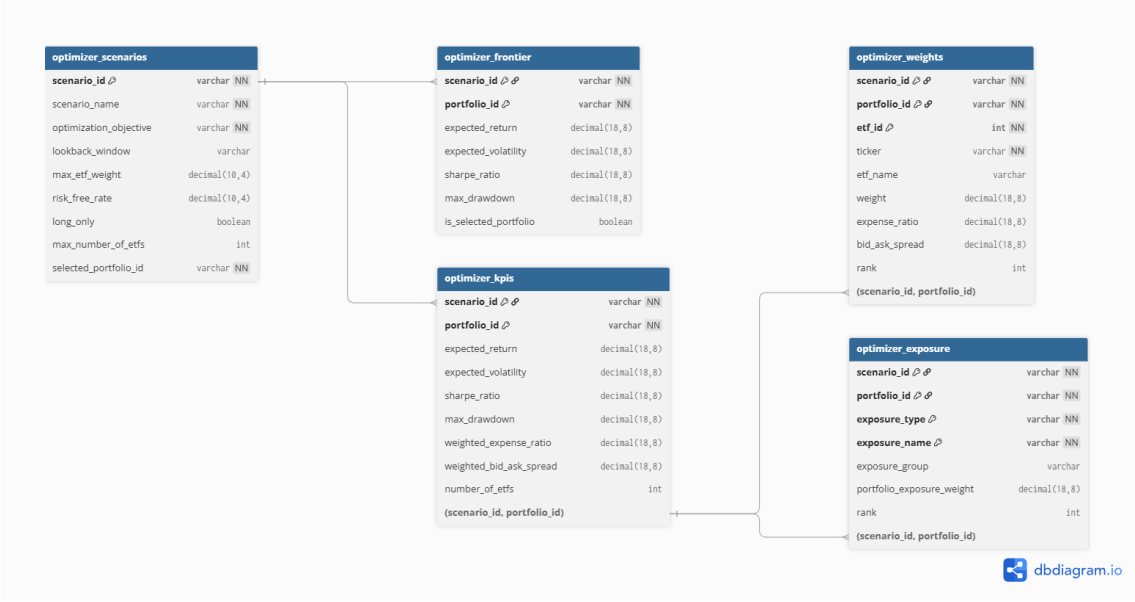


Figure 2: Optimizer schema for scenarios, efficient frontier, optimized weights, KPIs, and final exposure.

B Scenario 1 dashboard screenshots

This appendix presents the main Tableau Story scenario. It shows the full analytical path from ETF screening to final portfolio recommendation.

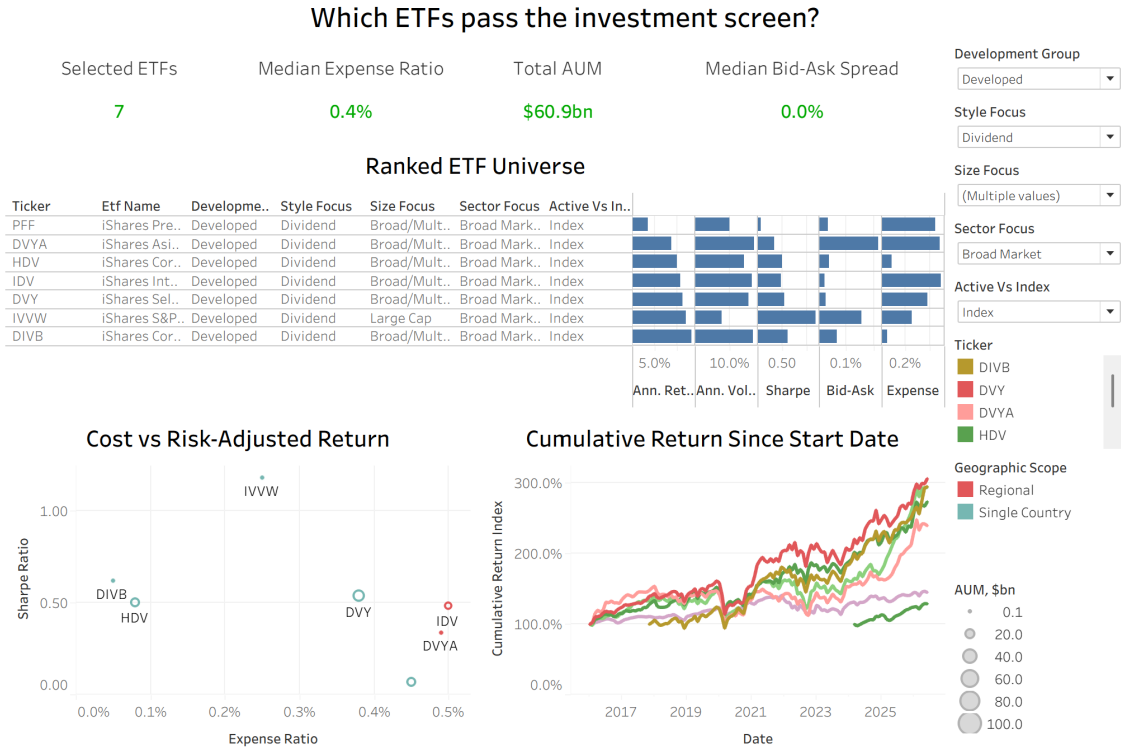


Figure 3: Scenario 1: ETF Screener dashboard.

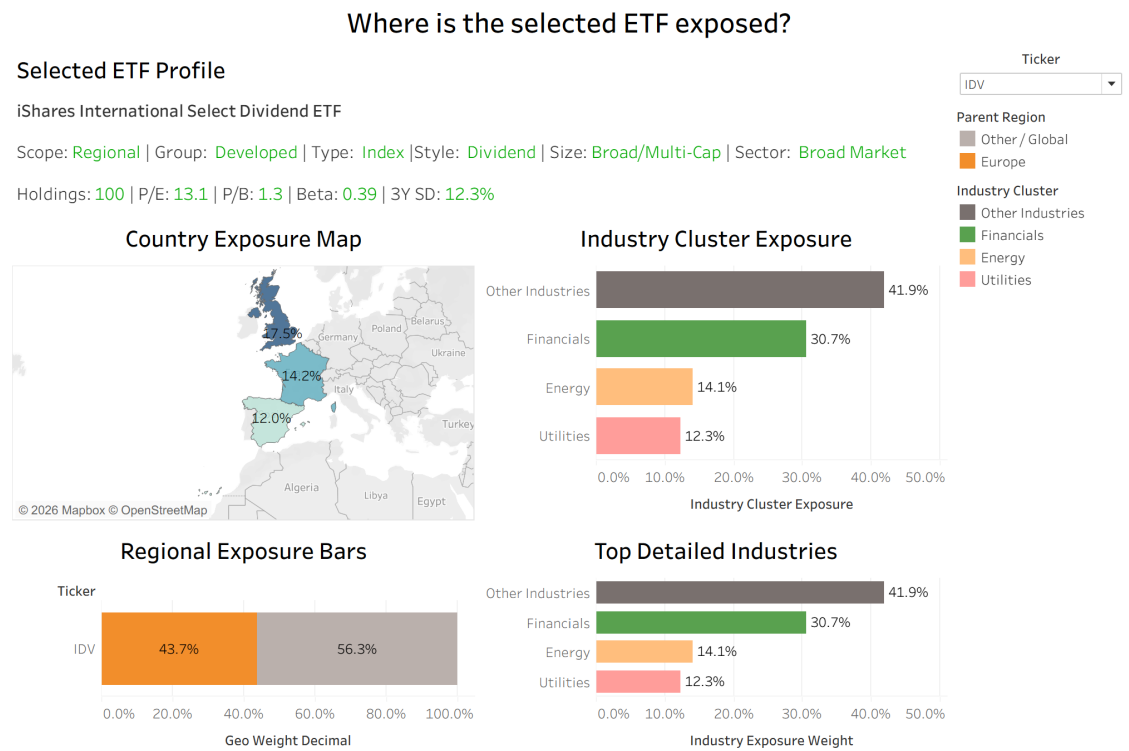


Figure 4: Scenario 1: ETF Exposure dashboard.

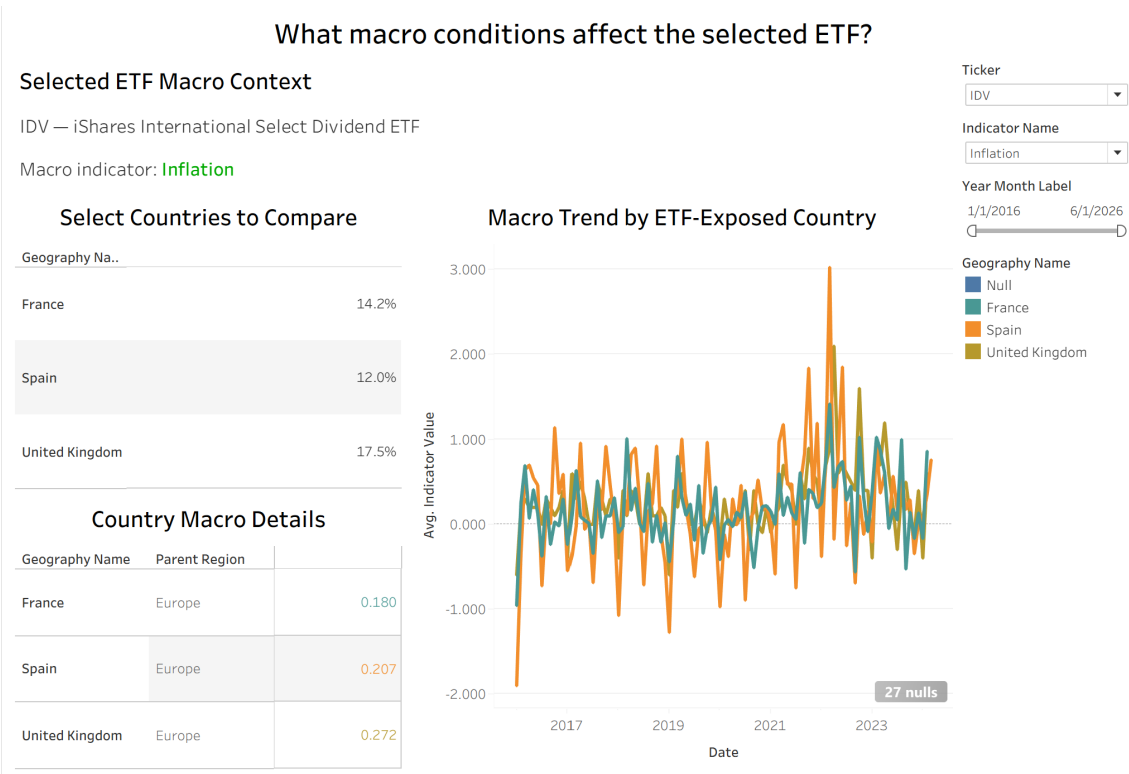


Figure 5: Scenario 1: Macro Context dashboard.

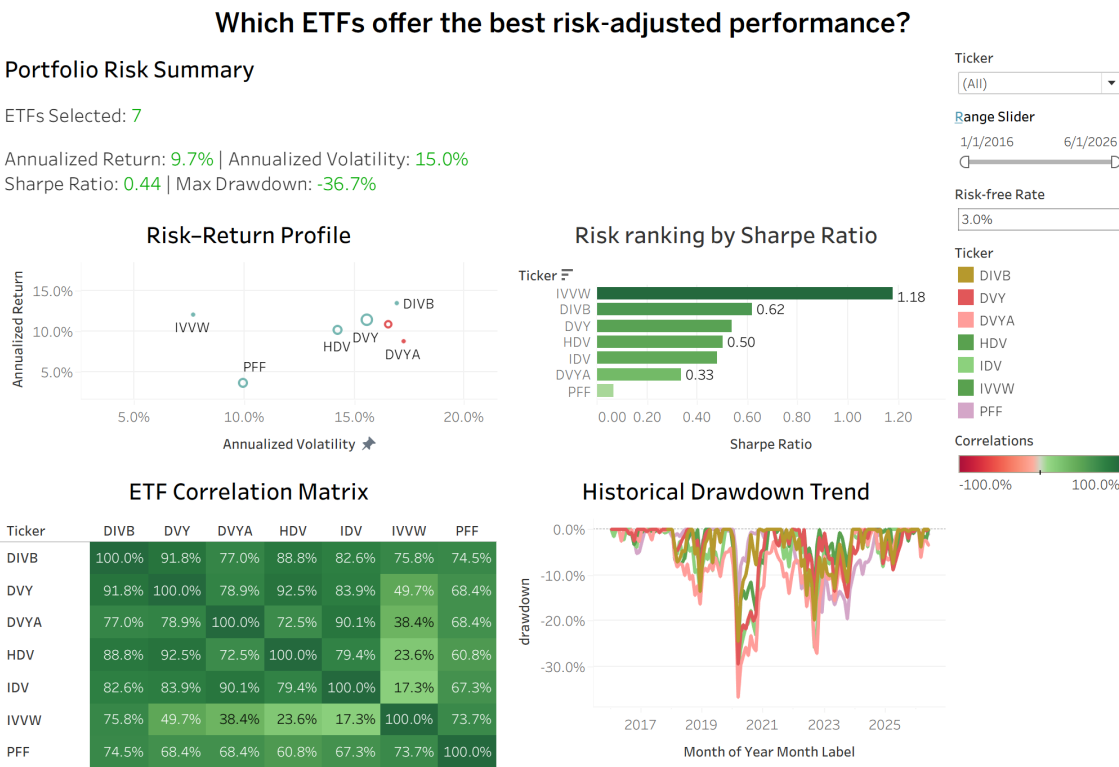


Figure 6: Scenario 1: Risk and Diversification dashboard.

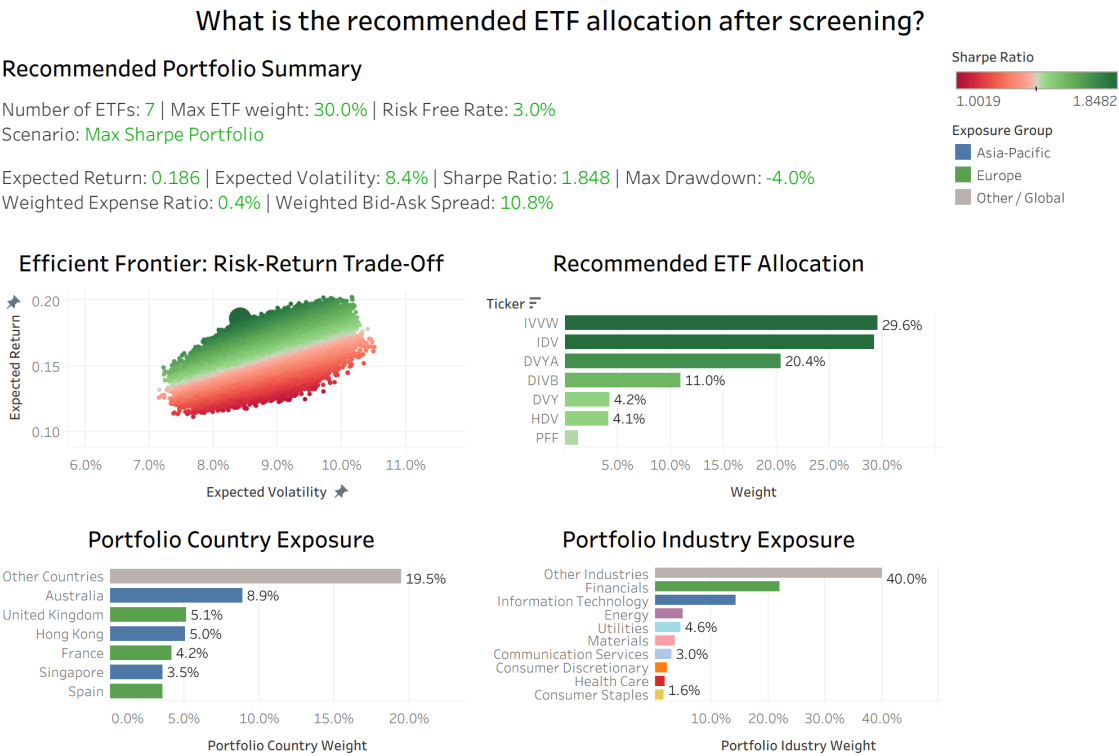


Figure 7: Scenario 1: Final Portfolio Recommendation dashboard.

C Scenario 2 dashboard screenshots

This appendix presents a second portfolio scenario. It demonstrates that the recommendation can change when the analyst changes the ETF selection or screening assumptions.

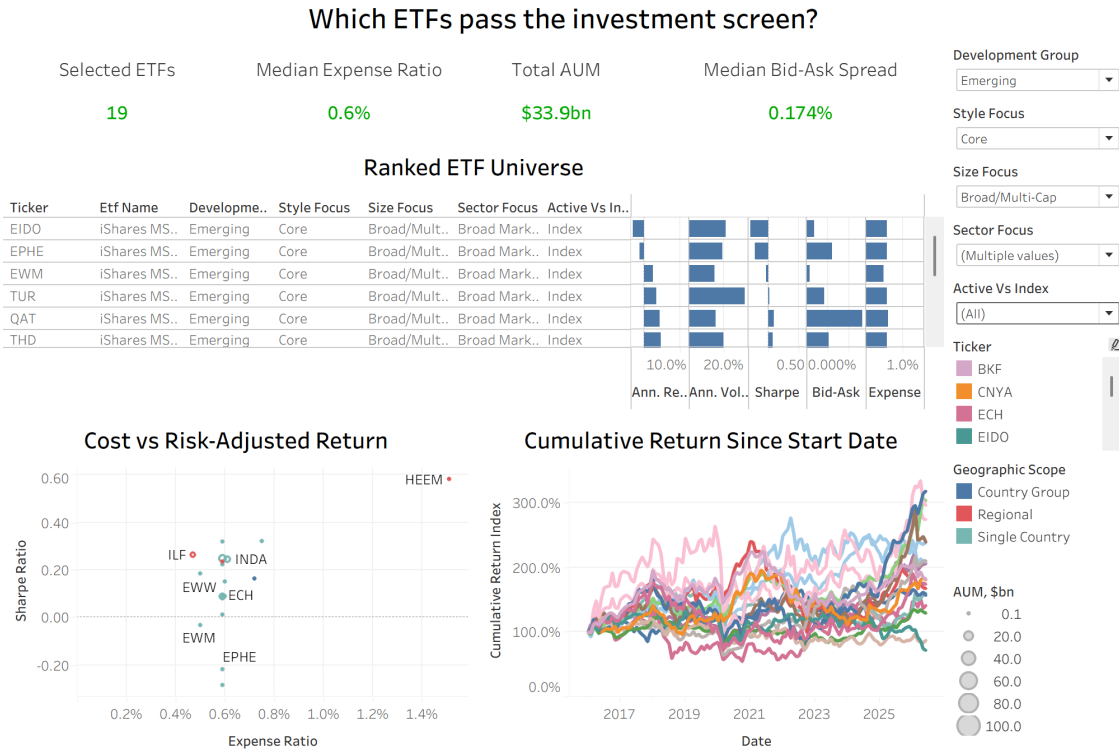


Figure 8: Scenario 2: ETF Screener dashboard.

What is the recommended ETF allocation after screening?

Recommended Portfolio Summary

Number of ETFs: 19 | Max ETF weight: 30.0% | Risk Free Rate: 2.0%
Scenario: Max Sharpe Portfolio

Expected Return: 0.093 | Expected Volatility: 13.7% | Sharpe Ratio: 0.5303 | Max Drawdown: -30.6%
Weighted Expense Ratio: 0.9% | Weighted Bid-Ask Spread: 20.0%

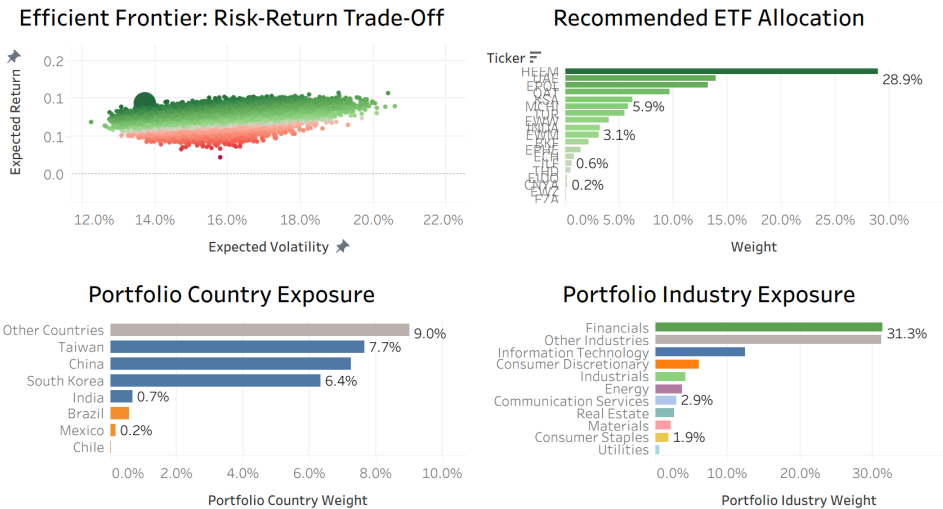
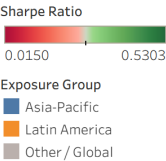


Figure 9: Scenario 2: Final Portfolio Recommendation dashboard.